IBM Delivery Central Platform Integration Playbook Ver 1.1

Table of Contents

1. Introduction....4
2. Available Integrations....4
2.1 Integrations with IBM Systems and Tools....4
2.2 Integrations with 3rd Party tools (Connectors)....5
3. Requesting a Connector to be configured for your Project....6
Appendix A....7
1 List of Available Connectors....7
2 IDCP and Tool's Field Mapping Details....7
3 Integration Request Form Template....7
4 Integrate Tools in Easy Steps....8
Appendix B - Technical Section....9
1 Integration Architecture....9
2 Integration Methods....10
2.1 Snaplogic....10
2.2 Import and Export....11
2.3 Adaptor....11
2.4 Embedding URL....12
2.5 IDCP Rest APIs....12
3 Network Connectivity Options....13
3.1 Integration over Private Network....13
3.2 Integration over Public Network....14
4 Security....15

Document History

Document Location

The IDCP Integration Playbook is also accessible from the following sites:

CSE w3 site > IBM Delivery Central Platform > Key Resources
IDCP Integration Playbook

Authors & Subject Matter Expert Contributors

The authors, subject matter experts (SME) and key contributors who have provided the content and reviewed the document for alignment and accuracy are listed below.

Name	Email ID	Role
Anne Netto	nettoa@sg.ibm.com	Lead Author & IDCP Integration PMG Lead
Jasbir Singh	jasbir.singh1@ibm.com	Co-Author – Architecture & Technical
Sireesha Namala	snamala1@in.ibm.com	Co-Author – User Integration Playbook
Luis Munoz	luis.munoz@us.ibm.com	Co-Author – IBM Consulting Essentials
Karaka C Kishore	kkaraka@in.ibm.com	IDCP IBM Integration Lead
Rajesh Rao	rrao@digite.com	IDCP Digite/Nimble Integration Lead
Amey Patil	apatil@digite.com	Integration SME – Digite/Nimble
Swapnil Bidwai	sbidwai@nimble.com	Integration SME – Digite/Nimble Snaplogic
Debapratim Chakraborty	debaprac@in.ibm.com	Integration SME – Platform Svcs (PESD)
Charu Malhotra	Charu.Malhotra1@ibm.com	IDCP Enablement Lead & SME

Version History

Changes to this document are summarized in the following table in reverse chronological order (latest version first).

Version Number	Revision Date	Summary of Changes	Changes marked
1.1	04-07-2024	IDCP Architecture Diagram updated	
1.0	01-08-2023	Initial version for publication	Changes Accepted
0.1	04-05-2023	Draft version distributed to IDCP program team &	Yes

Approvals

The following persons have approved this document:

Name and Email ID	Title	Approved Version
Aman Kumar Singhal	IDCP Program Director	1.0
Venkatesh Vittoba Kanna	IDCP Program Manager	1.0

1. Introduction

IBM Delivery Central Platform (IDCP) is IBM Consulting's new end to end delivery management and execution platform. IDCP transforms the way we deliver, by digitizing delivery workflows, embedding IBM assets into them and enabling real time governance, thereby bringing significant productivity gains. It simplifies and standardizes our delivery while reducing project losses. This platform, on IBM Consulting Essentials, will allow us to speed up the program launch, execution and enable deep collaboration across clients, partners and IBM.

One of the key strengths of this platform is its **configurable connectors** with 3rd party ALM (Application Lifecycle Management) tools like Jira, ServiceNow, Jenkins, GitLab and several others to help our PMs and practitioners work seamlessly. In addition, the joint IBM and Nimble (Digite) team has built **custom integrations** with some of our key IBM systems such as Integrated Project Planning & Forecasting (IPPF) data via Services Operations (SO) Hub, Professional Marketplace (ProM) and Project Database (GIC Project Db).

This document provides step by step guidance to any Program Manager, Project Manager and Architect to **request**, **provision** and **configure** these connectors for use within their project.

2. Available Integrations

2.1 Integrations with IBM Systems and Tools

IDCP offers some custom integrations with upstream IBM systems and tools that are trusted data sources of essential contract, program, project and team information related to the IDCP Program / Project. IDCP is also integrated with IBM Consulting Essentials to offer teams proven reusable services, assets & accelerators to deliver innovation and business value with speed. These custom integrations are jointly built by IBM and Nimble (Digite) as part of the foundational base product configuration and are **not modifiable at project level**. Other upstream and downstream integrations are being pursued to better integrate IDCP with IBM's enterprise systems and support intelligent workflows.

A list of the foundational base system integrations available at present is provided in the table below and will be updated to reflect any changes.

Integration	Purpose
Integrated Project Planning and Forecasting (IPPF)	To provide and maintain Commercial Project information such as Contract+Work No and essential contract / project attributes related to an IDCP Project when a Commercial Project is registered or updated in IPPF.

Project Database (GIC Project Db)	To provide and maintain CIC Project information related to an IDCP Program or Project when a CIC Project is registered or updated in Project Db. The CIC Project information is also updated when a Commercial Project based on Contract+Work No is registered or updated in IPPF and has an associated project in Project Db for the CIC scope of services.
Professional Marketplace (ProM)	To provide and maintain Team Members assigned to an IDCP Project when a project is setup and reflect any subsequent changes based on Professional Marketplace (ProM).
IBM Consulting Essentials	To enable invites and manage access of any external (non-IBM) users to collaboratively work in IDCP. To provide project teams with proven reusable IBM Consulting services, assets, and accelerators. To offer and provision service line / practice recommended recipes consisting of services and assets to project teams.
IBM Access Hub	To manage and grant IDCP access to IBMers with non project roles such as line of sight and support roles. (eg Market Leader, Service Line / Practice leader, IDCP AMS Support, IDCP PMG Leads, CSE Consultant, etc)

2.2 Integrations with 3rd Party tools (Connectors)

The second type of integrations are called connectors that are configurable for each project, and these typically enable integration with 3rd party tools. Here is a list of such connectors that are currently available. These could be **configured to work in specific project environments**.

As of Release 3.0 of IDCP

Tools	Tool Type	Data Flow	Data Flow Frequency	Integrated Features
Jira (Standard Config)	3rd Party	Tool->IDCP	5 mins – 24 hrs	Details of Sprint, US, Bugs, Epics, Release flow from Jira to IDCP
Jira (Practice Specific Config – HCT Migrate)	3rd Party	Tool->IDCP IDCP->Tool	5 mins – 24 hrs	Details of Sprint, US, Bugs, Epics, Release, ToDo flow from Jira to IDCP
Jira (Practice Specific Config - SAP)	3rd Party	Tool->IDCP	5 mins – 24 hrs	Details of Sprint, US, Bugs, Epics, Release, RAID, Epic Deliverables flow from Jira to IDCP
Jira (Practice Specific Config - SFDC)	3rd Party	Tool->IDCP IDCP->Tool	5 mins – 24 hrs	Details of Sprint, US, Bugs, Epics, Release, Features, ToDo flow from Jira to IDCP
Jira (Practice Specific Config – ED&M)	3rd Party	Tool->IDCP	5 mins – 24 hrs	Details of Sprint, US, Bugs, Epics, Release, Component, Project, Test, Improvement, Features flow from Jira to IDCP
Service Now (Standard Config)	3rd Party	Tool->IDCP	5 mins – 24 hrs	Summary of tickets, incident, service request and change request by priority flows from Snow to IDCP
MWS (Standard Config)	IBM Internal	Tool->IDCP	5 mins – 24 hrs	Export and import of all the required parameter to IDCP Eform.
Jenkins (Standard Config)	3rd Party	Tool->IDCP	On Commit	Build details captured in build Eform instance.
Gitlab (Standard Config)	3rd Party	Tool->IDCP	On Commit	All commit (commit against any User Story, BUG etc) flows to IDCP GIT Eform instance.

Refer IDCP Available Connectors for the most up to date list.

3. Requesting a Connector to be configured for your Project

Project Managers who wish to integrate a 3rd party tool to their IDCP Project may request and provide the requirements in an Integration Request Form (IRF).

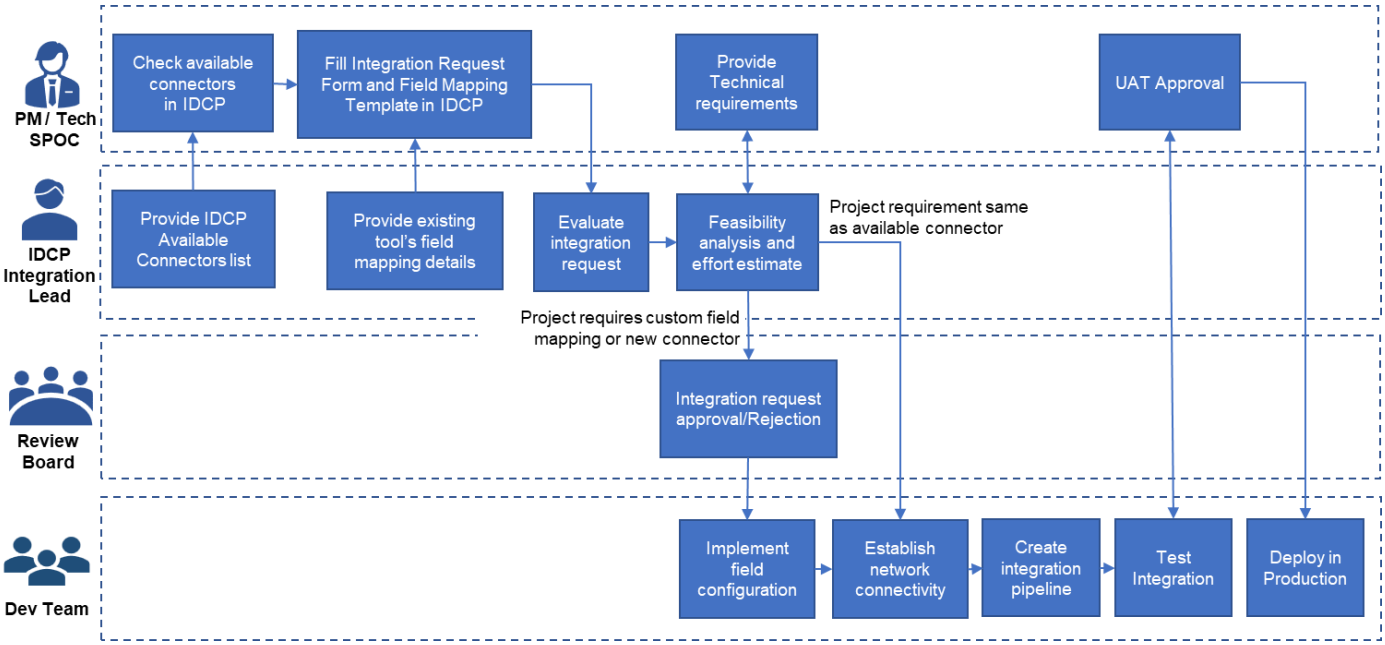
There are 4 standard scenarios of configuring the connectors available in IDCP catalogue.

Scenario	Target tool in predefined environment (e.g., Jira in IBM Liberty)	Standard Field mapping (No changes to predefined mapping of fields)
1	Yes	Yes
2	No	Yes
3	Yes	No
4	No	No

The IDCP Integration Team Lead from Nimble (Digite) will review the request and provide the estimate for connector configuration based on the above scenarios.

In some cases, projects may require an integration with a 3rd party tool that is NOT in the above catalogue of connectors, although used extensively by client in co-executed delivery. In such cases, the IRF will go through the Integration Change Control Board for approval. These requests may typically involve a longer lead time for the development.

The below diagram explains the process of how an IRF request would be handled.



Note: There are plans to automate the management of Integration Requests (IRFs) within IDCP in future.

Refer IDCP and Tool Field Mapping for details on standard field mappings for existing connectors.

Appendix A

1 List of Available Connectors

Please refer to "IDCP Available Connectors" document for updated list of connectors.

https://ibm.box.com/s/t6mq6us3ryd6tf24rhm5zu9okgyaggl2

2 IDCP and Tool's Field Mapping Details

1. Jira
   https://ibm.box.com/s/497mm7y3qz24jtugn2kiwjagfbg2ip99

2. Service Now
   https://ibm.box.com/s/yb335lazlmn8zoa98qtunejrrni6ss4p

3. Jenkins
   https://ibm.box.com/s/n7re73acz5ve09e5484rimf8jn2juzag

4. Gitlab
   https://ibm.box.com/s/1d4q0mb8obw26hqhtw4gxxldff7trihs

3 Integration Request Form Template

The Project Manager must submit a request for any integration required with IDCP.

The Integration Request Form template and submission details is available in Box at:
https://ibm.box.com/s/mysrzgfqcs1lycbmyec0jttnww43kgn1

Note: There are plans to automate the management of Integration Requests (IRFs) within IDCP in future.

4 Integrate Tools in Easy Steps

Step 1: Provide Source System instance details

1. Location of Source System instance – IBM enterprise network or Client network
2. Whether required REST APIs are enabled for integration
3. Required access credentials to invoke the REST APIs

Step 2: Create connectivity between Source System instance and IDCP

Source System in IBM network: Project team must work with Source System Admin to create firewall rules to allow connectivity between Source System instance and IDCP server.

Source System in Client network: Project team must work with Client to establish connectivity from IDCP to their Source System instance.

Step 3: Check if project requirements met by the IDCP standard Source System field mapping

If project requires same field mapping as of IDCP standard field mappings of Source System (Refer IDCP and Tool's Field Mapping Details) then enable required APIs of Source System instance.

If changes are required in IDCP standard filed mappings of Source System. The project team must work with the Source System Admin to get the details of the exact APIs to be invoked for the requirements.

Step 4: PM to create Integration Request in IDCP

The Technical SPOC from the project must provide the details of Source System instance and attach the mapping of Source System fields to IDCP fields in integration request (Refer Integration Request Template).

Step 5: Review of Integration Request

If custom changes to the Standard field mappings is required, the Integration request will be reviewed by Integration team.

Step 6: Configuration and Deployment

If standard field mapping applies, the configuration and deployment would take an estimated 2 weeks. For any customization, the Integration team will provide the estimated effort and timelines for deployment.

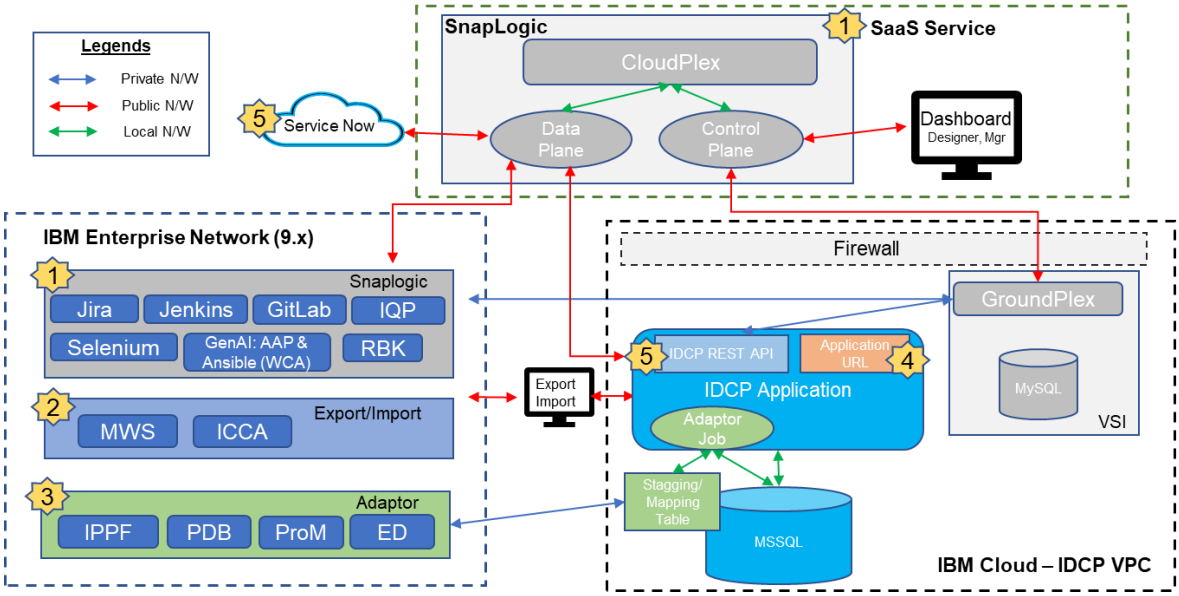
Appendix B - Technical Section

1 Integration Architecture

IDCP has multiple methods of integration with IBM system, IBM internal application, 3rd Party applications and client's tools or applications. Integration data will flow over private and secure (IPsec tunnel) network for IBM system and IBM internal tools and applications. For 3rd party and Client's applications and tools integration, data can flow either over private network (site to site VPN) or over public network.

Following are four integration methods IDCP supports: -

1. Snaplogic
2. Import and Export
3. Adaptor
4. Embedding URL of Applications or Tools
5. IDCP Rest APIs



2 Integration Methods

2.1 Snaplogic

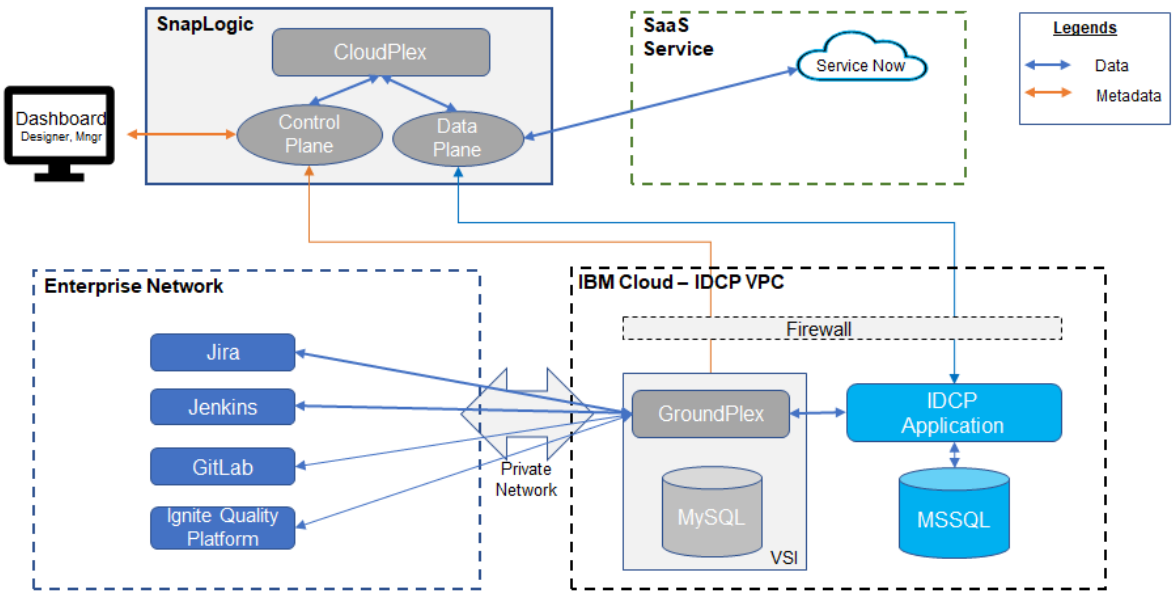
The Snaplogic GUI connects to the Control Pane through REST API via the HTTP protocol which orchestrate the integrations and a Data Pane (Cloudplex or GroundPlex) which does all the transformations and heavy lifting activities. The Cloudplex and GroundPlex connect to the application and transfer data. The Snaplogic Cloud has mainly three parts - Snaplogic Designer, Snaplogic Manager, Snaplogic Dashboard. Through the Snaplogic Designer the user can Build, Configure and Execute a Pipeline. All activities in Snaplogic are managed by Nimble (Digite) SMEs.

Snaplogic encapsulates integration task complexity with “Snaps”. Snap can perform as simple a task as reading data from a file, or as complex an operation as connecting to an instance of Salesforce.com, analysing the source data, and providing full access (data and functionality) to all standard and custom objects within Salesforce.com.

Snaplogic GroundPlex instance is deployed in IDCP infrastructure which provide private and secure data integration with applications. IDCP support only REST API based integrations. Thus, applications or tools must have REST APIs support available to integrate it with IDCP via Snaplogic.

Refer Snaplogic details in this document:

https://ibm.box.com/s/kmsujt3zy0odepdovxe56sqtur09b7x4



2.2 Import and Export

IDCP also supports importing and exporting data between IDCP and external applications in multiple formats.

Exporting work items enables sharing files and helps in adding and modifying work items offline using MS Excel. Users may export the structure to add and upload items or all content for all fields to modify and/or close items. On Importing the files back into the application, items are added/updated/closed as appropriate.

To bring in work items from other systems into the application, users may export the structure of that work item, i.e. Skeletal Export, and then enter data formatted to the columns in the file before importing the file into the application.

Refer to IDCP import and export details in this document:
https://ibm.box.com/s/40dfsu6bpqk5cofmssswwfyq7ci3eg9h

2.3 Adaptor

Adaptor Framework is an integration framework that facilitates integration of entities between IDCP and External Systems. Adaptor framework comprises of Staging area, for populating data from external systems, and Transformation logic for data transform. An Adaptor is registered in the IDCP Database.

The Staging area consists of two types of tables:

Staging Tables

Staging tables are provided as part of the IDCP database as an interface to the external systems to provide their data. External Adaptors can be designed to fetch data from the external systems and populate the Staging. Internal adaptors provided by IDCP can then pick data from these temporary storage tables and complete the synchronization.

Mapping Tables

They are Data Definition or Data Mapping tables. The staging tables are provided with columns to set mandatory details that are required for creating/editing/etc of an entity. They have meaningful names to match the corresponding Nimble (Digite) attribute. Additionally, the staging table is provided with general fields that can be used to map custom or other Nimble attributes. This mapping is stored in mapping tables.

Once data is inserted into staging table, REST API needs to be triggered with required Job Name & runID to execute the job.

https://ibm.digite.com/rest/v2/#/IntegrationService/ExecuteJob

Refer IDCP adaptor base integration details in this document:

https://ibm.box.com/s/shh6tce2bp1jpi8kpn6cqslmgunb6dxz

2.4 Embedding URL

URLs to applications and tools may be embedded in IDCP or any of its Process Templates or functions for quick access. This is not an integration, and no feed or exchange of data is involved. However, this may be useful in situations where ease of access to specific applications or tools from IDCP is required to enhance the users experience.

Note that access to these applications must be requested and obtained separately.

2.5 IDCP Rest APIs

You can use Nimble (Digite) REST APIs as service endpoints to interact with IDCP remotely and perform various functions.

Nimble REST web service exposes API to allow users to perform common card operations as listed below.

- CalenderService
- MasterListService
- ECRService
- TokenService
- EFormService
- BoardService
- OrganizationService
- UserService
- IntegrationService
- ProjectService
- TeamService

- TaskService
- ToDoService
- TimeSheetService

Refer to the complete list of Nimble (Digite) REST APIs in this document:

https://ibm.box.com/s/rlzdgf2oitevji7al0jnc3w28abuub97

3 Network Connectivity Options

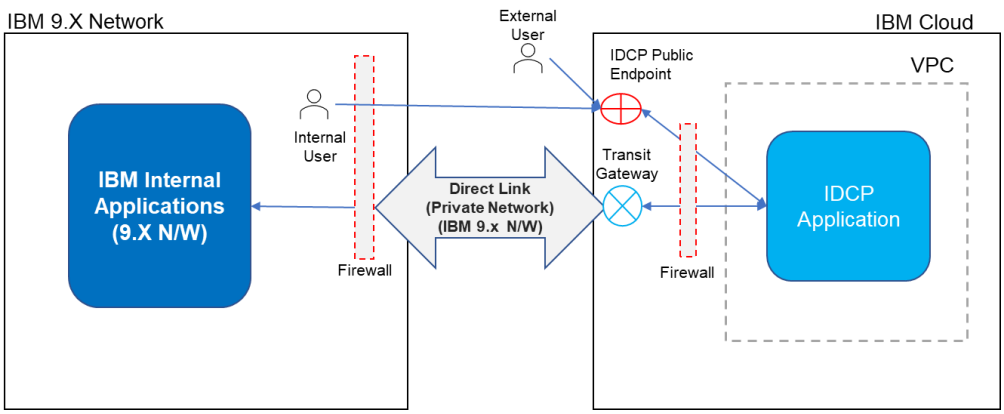
There are two network connectivity options available for integration, depending on the specific needs of the integration project.

3.1 Integration over Private Network

To establish connectivity to application and tools which are hosted on premises and enterprise network.

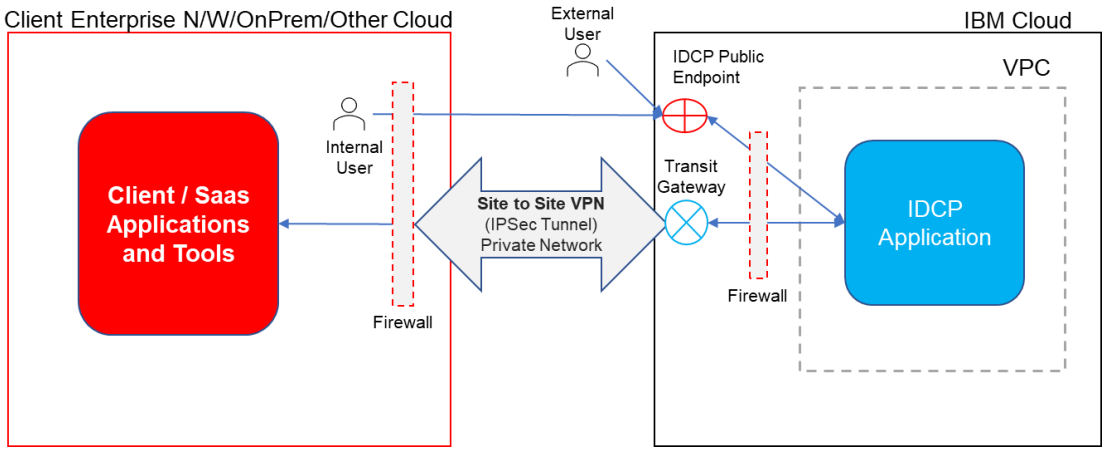
1. Direct Link

A Direct Link is provided by IDCP to IBM Enterprise network (9.x) that allows IBM internal applications and tools to establish a dedicated, private connection. Direct Link can be used to bypass the public internet and establish a secure, high-bandwidth connection.



2. Site to Site VPN

Site to Site VPN (Virtual Private Network) provides a secure way to connect two or more networks together over the internet. This can be useful in situations where data needs to be exchanged between systems in different locations, and security is a concern.

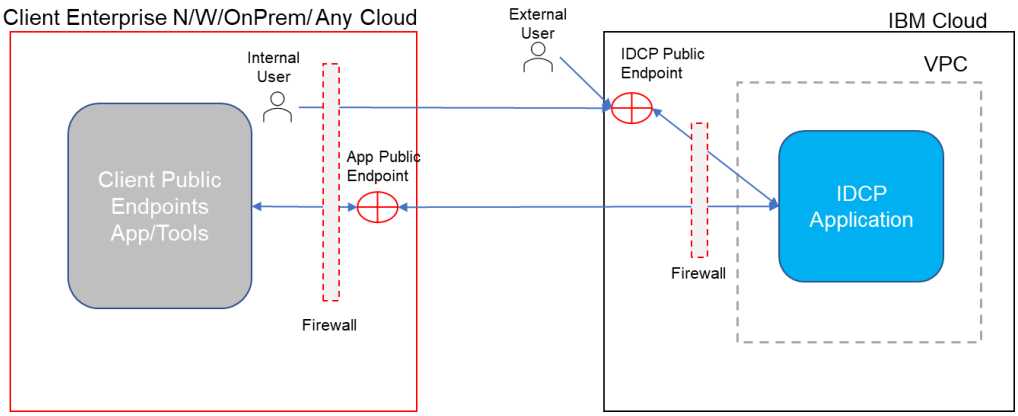


3.2 Integration over Public Network

To establish connectivity to application and tools hosted on public network or have public endpoints for Rest API available.

1. Firewall Rules

To establish connectivity to applications or tools exposed to public endpoints for Rest APIs in a secure way, whitelisting of applications or tools, IP addresses or domain name and ports can be applied on IDCP firewall. Similarly, IDCP application IP addresses or domain name and port can be whitelisted in Client's firewall.



4 Security

IDCP infrastructure is hosted on IBM Cloud which include CIS with DDOS protection and WAF, FortiGate Firewall and other security components integrated in the architecture.

IDCP web URLs are completely secured using TLS certificates and IBMid SSO authentication. End users should have an IBM User ID to login to IDCP.

IBM Delivery Central Platform (IDCP) is committed to protect the privacy and confidentiality of personal information about its Employees, Clients, Business Partners, and other identifiable individuals. Uniform practices for collecting, using, disclosing, storing, accessing, transferring, or otherwise processing such information assists IBM to process personal information fairly and appropriately, disclosing it or transferring it only under appropriate circumstances.

With IBM Cloud Security, we encrypt the data at rest, in transit, and in use, for any data storage type—block, files, databases, data services, and objects storage.

For further information, kindly refer to IBM Delivery Central Platform Security Point of View.